Layal Canoe

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Portfolio | GitHub

Education

Effat University — Jeddah, Saudi Arabia
Bachelor of Computer Science – Artificial Intelligence — *2022 – 2026*

- CGPA: 3.97/4.00
- Dean's List 2022–2023, 2023–2024, and 2024–2025

Experience

AI Intern — Jun. 2026 – Present
Heroz — *Jeddah, Saudi Arabia*

- Built a hybrid recommendation system for personalized children's activity discovery using interests, preferences, developmental goals, behavioral signals, exploration, and session suitability.
- Designed MongoDB data models and a Neo4j Knowledge Graph mapping children, activities, interests, goals, learning outcomes, and behavioral evidence.
- Developed graph-building and multi-hop traversal logic using JavaScript, Node.js, Neo4j, and Cypher to discover recommendation candidates.
- Designed a 6-factor weighted scoring model with eligibility filtering, missing-data normalization, ranking logic, and explainable recommendation evidence.

Back-End Developer Intern — Summer 2025
Breifiction – Marketing Agency — *Jeddah, Saudi Arabia*

- Built 5+ REST API endpoints powering a personalized AI assistant for customer messaging, shipment tracking, and order-status responses.
- Integrated AI functionality with client databases and designed SQL/NoSQL schemas supporting real-time, context-aware data retrieval.
- Implemented authentication and backend security for production-oriented services.

Research & Publications

Scalable Multi-Agent Edge Intelligence for Traffic Coordination | *NLP & Big Data*

- Engineered a distributed AI pipeline using Kafka, Hadoop HDFS, Spark/PySpark, and Spark MLlib across 8 Jeddah intersections, benchmarking 1,000,000 sensor records and improving throughput from 171 to 13,791 records/sec (80.6×).
- Developed a TF-IDF + Logistic Regression severity classifier achieving 100% accuracy on clean data and 84.9% under 20% adversarial sensor dropout.
- Integrated Vehicle, Incident Detection, and Traffic Light agents with RAG and Knowledge Graph reasoning for downstream traffic decision-making.

IEEE Conference Publication | *AI & Cybersecurity in Powerline Communication Systems*

- Co-authored peer-reviewed research accepted at the IEEE SCC Conference in Sousse, Tunisia and selected for research funding by Effat University's Deanship of Graduate Studies and Research.
- Investigated AI-enabled cybersecurity for smart-grid and IoT infrastructure, including anomaly detection and post-quantum cryptography for Powerline Communication systems.

Selected Projects

AgentNation – AI Policy Simulator | *Python, Multi-Agent Systems, RAG, Knowledge Graphs*

- Designed a four-layer multi-agent architecture for cross-ministry policy simulation, integrating specialized agents for policy analysis, workforce intelligence, economic impact, and cross-policy reasoning.
- Built a functional Python prototype combining deterministic scoring, RAG, Knowledge Graph context, and human-in-the-loop decision support.

Hajj Buddy AI | *RAG, NLP, LLMs, Node.js*

- Built a RAG-based assistant for Hajj and Umrah that grounds responses in retrieved content from verified Islamic sources.

- Designed the system to decline when supporting context is insufficient and deployed a live conversational interface for source-grounded responses.

IoT Intrusion Detection System | *Python, Machine Learning, Deep Learning*

- Built an intrusion-detection pipeline on approximately 200,000 CIC IoT 2023 network samples, applying SMOTE for class balancing and PCA for dimensionality reduction.
- Benchmarked Random Forest, XGBoost, MLP, and CNN models, with Random Forest achieving the highest accuracy at 95%.

Music Map – Full-Stack Music Discovery Platform | *JavaScript, PHP, SQL, HTML/CSS*

- Developed a full-stack music discovery platform with a JavaScript frontend and PHP/SQL backend supporting CRUD operations, authentication, and database-driven content.

Technical Skills

Languages: Python, JavaScript, Java, C++, SQL, PHP, Cypher, HTML/CSS
AI & Machine Learning: Machine Learning, Deep Learning, RAG, LLM Integration, NLP, AI Agents, Multi-Agent Systems, Knowledge Graphs, Computer Vision
Data & Big Data: MongoDB, Neo4j, Apache Spark/PySpark, Kafka, Hadoop HDFS, Spark MLlib, Pandas, NumPy
Backend & Automation: Node.js, REST APIs, SQL/NoSQL Databases, n8n, Workflow Automation
Developer Tools: Git/GitHub, Docker, Linux, Postman, Streamlit

Honors & Awards

- DGRS Research Funding Award – Effat University
- KAUST Academy AI Specialization – Stage 2 Selection, National Competitive Assessment, 2025